

**A
PROJECT REPORT
ON
ONLINE BUS BOOKING SYSTEM**



**Bachelor of Technology
In
Computer Science & Engineering
By
Vishal Chaudhary**



**SCHOOL OF COMPUTER SCIENCE & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE
GOVT. P.G. COLLEGE DHARAMSHALA
Batch 2022 – 26**

AN
INDUSTRIAL TRAINING REPORT
**ONLINE BUS
BOOKING SYSTEM**

Submitted in partial fulfillment of the Requirements.
For the award of Bachelor of Technology in Computer Science and Engineering

**HIMACHAL PRADESH TECHNICAL UNIVERSITY
HAMIRPUR (H.P.) - INDIA**



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CERTIFICATE

This is to certify that the work embodied in this dissertation entitled “**ONLINE BUS BOOKING SYSTEM**” has been satisfactorily completed by **Vishal Chaudhary (23020603008)**. It is a bonafide piece of work, carried out under our / my guidance in the **Bachelor of Technology, Govt. P.G. College, and Dharamshala** for the partial fulfillment of the **Bachelor of Technology** degree during the academic year 2024-25.

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representing approximately 300 hours of work

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DECLARATION

I, **Vishal Chaudhary (23020603008)**, a student of **Bachelor of Technology, Govt. P.G. College, Dharamshala** hereby declare that the work presented in this Project is outcome of my own work, is bonafide, correct to the best of my knowledge and this work has been carried out taking care of Software Ethics. The work presented does not infringe any patented work and has not been submitted to any University for the award of any degree or any professional diploma.

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ABSTRACT

The project: “Online Bus Booking System” is a web-based platform designed to simplify the process of booking bus tickets and manage travel schedule. It enables users to browse available bus routes, check seat availability, compare fares, and make reservations in real-time from any location. The system streamlines operations for both passengers and bus operators by providing features such as user registration, secure payment gateways, automated ticket generation, and instant booking confirmations. Bus operators can manage schedules, monitor bookings, and update seat availability with ease. The system enhances the overall efficiency of travel planning, reduces the need for physical ticket counters, and provides a convenient, user-friendly interface for passengers. This solution aims to improve customer experience, reduce human error, and increase operational efficiency in the bus transport sector.

For bus operators, the system simplifies the management of schedules, seat allotments, and bookings through a centralized interface. This enhances operational efficiency, reduces the chances of human error, and allows for quick updates to schedules or seat availability. By automating various processes, the system improves both customer experience and the internal operations of transport companies, making bus travel more accessible and user-friendly.

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1. Introduction

1.1 Problem Statement

In this Covid-19 situation, bus owners and passengers have faced the problem where only limited number of passengers are allowed to travel on a single bus. It would be more convenient if passengers can book their seat tickets from home. So, owners also can get number of passengers to the exact number of seats to perform their service without breaking the guidelines. Also, in case of a Covid 19 patient is recognized, buses need to have a list of passengers who are traveling in the bus throughout the day. With the advancement of technology, the most convenient medium to fulfill these needs is the online platform.

1.2 Product Scope

Online ticket reservation system will be the best solution for the bus owners with the Covid-19 travel restrictions. This is a website which enables passengers to view a bus and book the ticket by their own. A passenger has given the chance to book his desired seat by the remaining seats. This website is developed with a clean interface where passengers can select the bus, view the operator, view destinations, and book a seat/s. Main benefit of the website will be saving time and helping in maintaining the social distancing among public in this pandemic.

1.3. Objectives of Study

The main purpose of this study is to automate the manual procedures of reserving a bus ticket for any journey made through Imo Transport Company (ITC). This system is said to be an automatic system and customers can select seats by themselves. Specifically, objectives of this project will consist of:

- i) Providing a web-based bus ticket reservation function where a customer can buy bus ticket through the online system without a need to queue up at the counter to purchase a bus ticket.
- ii) Enabling customers to check the availability and types of busses online. Customer can check the time departure for every ITC bus through the system.
- iii) Easing bus ticket payment by obtaining a bank pin after payments is made to the various designated banks.

- iv) Ability of customers to cancel their reservation.
- v) Admin user privileges in updating and cancelling payment, route, and vehicle records.

1.4 Project Report Structure

This report comprises of three parts respectively introduction, methodology and conclusion. In the methodology part includes the functional and non-functional requirements, respective use case diagrams, activity diagrams which explain the scenario, ER diagram and the interfaces of each page. In Conclusion part it is has been explained about the overall view and the need of the report along with the project results, lessons learned and the future work. In appendices part the some of the important code has been added.

2. Research Methodology

The system of collecting data for research project is known as research methodology. The data may be collected for either theoretical or practical research for example management research may be strategically conceptualized along with operational planning method and change management. Information which was used for this study was carried out by oral interview.

2.1 Methodology

2.1.1 Requirements and Analysis

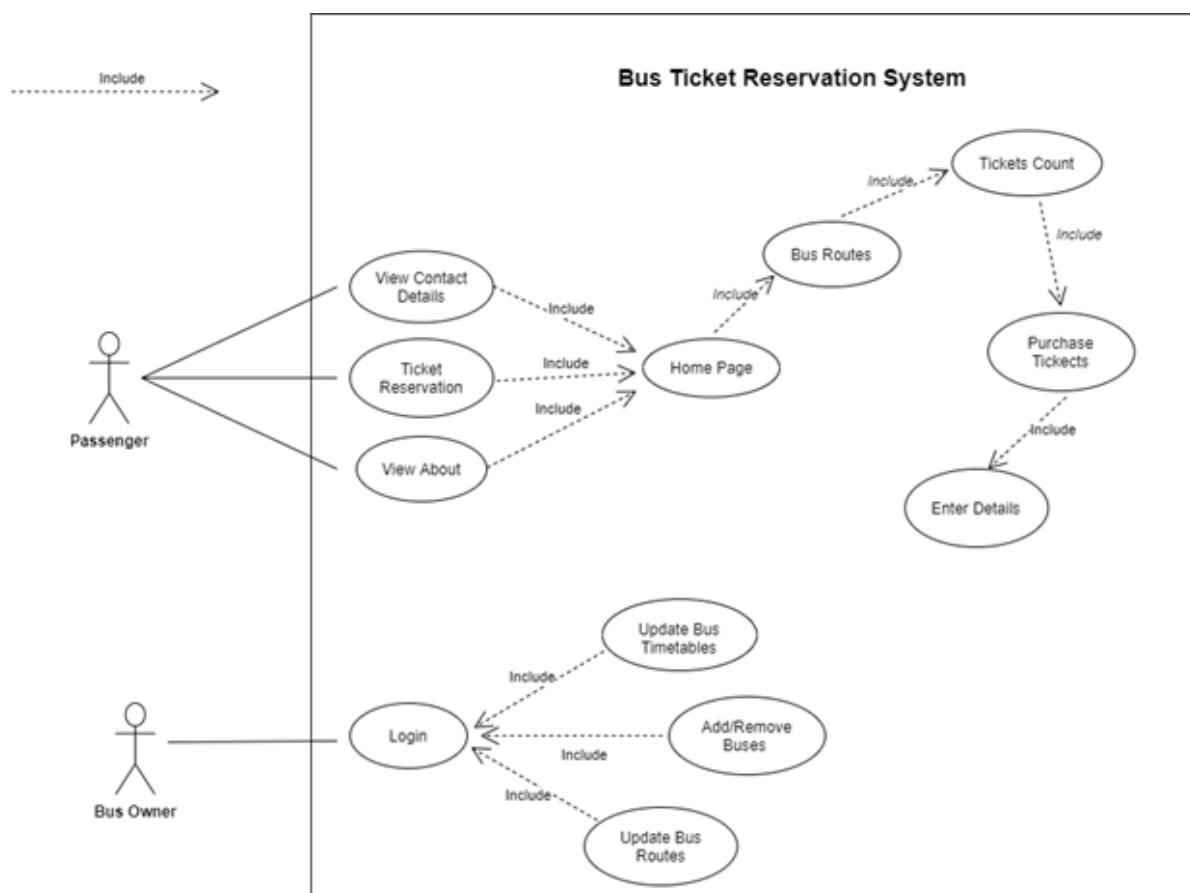


Figure 1 bus ticket reservation

2.2 Choice of Methodology

For any project to be completed, it must go through stages called Development Life Cycles. System Development Life Cycle (SDLC) is the process of understanding how an Information System (IS) can support business needs, designing the system, building it and delivering it to users. The SDLC composes of four phases: Planning, Analysis, Design, and Implementation.

2.2.2 Design Diagrams

Flow of the project

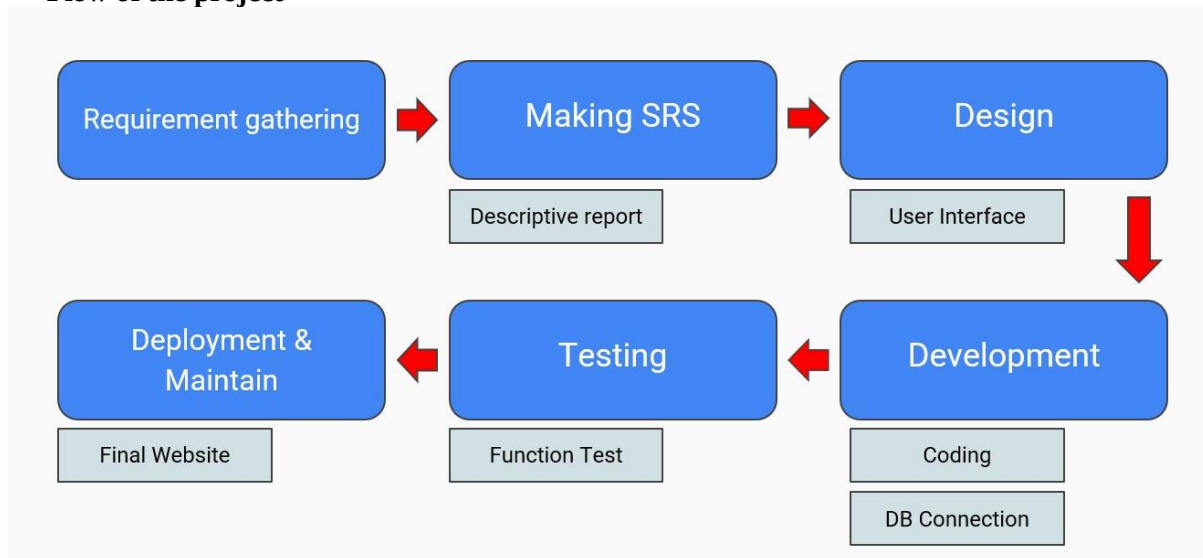


Figure 2 Design Diagram

2.3 About Online Bus Booking Management System

Online Bus Booking Management System contains of admin and user, the admin can manage the booking from the customer like monitor booking, add bus information, assign location, and the user is representing as a customer no need to create account just write information and copy the generate code after booking, and the admin can contact the code after booking customer. This simple bus booking system is developed in PHP MySQL database, using JavaScript, HTML, CSS, modal, Ajax, and bootstrap, the system is easy to use and simple to understand.

Features of the Project

- **Online Admin Dashboard**

Home, Schedule, Booklist, Maintenance, bust list, Location List, User, Manage Account.

- **Schedule**

Show Entries, Search, Id Numbers, Date, Bus, Location, Departure, ETA, Availability, Price, Action, Delete, and Edit.

- **Booked List**

ID Number, Ref Number, Full Name, Quantity, Amount, Status, Show Entries, Search, and Edit.

- **Bus List**

ID Number, Bus Number, Bus Name, Show Entries, Edit, Delete, and Search.

- **Location List**

ID Number, Terminal, City, Province State, Show Entries, Edit, Delete, and Search.

- **User**

ID Number, Name, Username, Edit, Delete, Show Entries, and Search.

- **User Dashboard Schedule**

ID Number, Date, Bus, Location, Departure, ETA, Availability, Price, Book, Show Entries, and Search.

2.4 Technologies Used

- **Html**

Hypertext: text (often with embeds such as images, too) that is organized to connect related items

Markup: a style guide for typesetting anything to be printed in hardcopy or soft copy format

Language: a language that a computer system understands and uses to interpret commands.

HTML determines the structure of web pages. This structure alone is not enough to make a web page look good and interactive. So, you'll use assisted technologies such as CSS and JavaScript to make your HTML beautiful and add interactivity, respectively.

- **CSS**

Cascading Style Sheets (CSS) is a powerful language used to enhance the presentation of HTML documents. CSS dictates how elements should be rendered across various media types, including on-screen, print, and speech. The cascading aspect of CSS means that styles can be applied not only to a specific element but also inherited by its children, creating a hierarchy of styling.

- **PHP**

The abbreviation PHP initially stood for Personal Homepage. But now it is a recursive acronym for Hypertext Pre-processor. (It's recursive in the sense that the first word itself is an abbreviation, so the full meaning doesn't follow the abbreviation.)

- **XAMPP Control Panel**

XAMPP is one of the widely used cross-platform web servers, which helps developers to create and test their programs on a local webserver. It was developed by the **Apache Friends**, and its native source code can be revised or modified by the audience. It consists of **Apache HTTP Server, MariaDB, and interpreter** for the different programming languages like PHP and Perl. It is available in 11 languages and supported by different platforms such as the IA-32 package of Windows & x64 package of macOS and Linux.

Components of XAMPP

As defined earlier, XAMPP is used to symbolize the classification of solutions for different technologies. It provides a base for testing of projects based on different technologies through a personal server. XAMPP is an abbreviated form of each alphabet representing each of its major components. This collection of software contains a web server named **Apache**, a database management system named **MariaDB** and scripting/ programming languages such as **PHP** and **Perl**. X denotes Cross-platform, which means that it can work on different platforms such as [Windows](#), [Linux](#), and macOS.

Many other components are also part of this collection of software and are explained below.

1. **Cross-Platform:** Different local systems have different configurations of operating systems installed in it. The component of cross-platform has been included to increase the utility and audience for this package of Apache distributions. It supports various platforms such as packages of Windows, Linus, and MAC OS.
2. **Apache:** It is an HTTP a cross-platform web server. It is used worldwide for delivering web content. The server application has made free for installation and used for the community of developers under the aegis of Apache Software Foundation. The remote server of Apache delivers the requested files, images, and other documents to the user.
3. **MariaDB:** Originally, MySQL DBMS was a part of XAMPP, but now it has been replaced by MariaDB. It is one of the most widely used relational DBMS, developed by MySQL. It offers online services of data storage, manipulation, retrieval, arrangement, and deletion.
4. **PHP:** It is the backend scripting language primarily used for web development. PHP allows users to create dynamic websites and applications. It can be installed on every platform and supports a variety of database management systems. It was implemented using C language. PHP stands for **Hypertext Processor**. It is said to be derived from Personal Home Page tools, which explains its simplicity and functionality.
5. **Perl:** It is a combination of two high-level dynamic languages, namely Perl 5 and Perl 6. Perl can be applied for finding solutions for problems based on system administration, web development, and networking. Perl allows its users to program dynamic web applications. It is very flexible and robust.

3. Software Requirement Specification Document (SRS)

3.1 Functional Requirements

3.1.1 Passenger side Requirements

In the homepage contains a brief description about the Online bus ticket reservation system such as bus details contact numbers and driver details.

Priority level: High

REQ-1	Enter the Website through browser
Input	Click the URL
Process	Loading system main page
Output	Accessing the homepage
Definition	Display the Details of Webpage

REQ-2	View Bus Details
Input	Click the Bus details button
Process	Loading the bus detail page
Output	Showing Bus Details
Definition	Display the Details of Available busses

REQ-3	View Driver's Details
Input	Click the Our Driver details button
Process	Loading the Driver details page
Output	Showing Driver's Details
Definition	Display the Details of Drivers

REQ-4	View Contact Details
Input	Click the Contact details button
Process	Loading the Contact details page
Output	Showing Contact Details
Definition	Display the Contact Details

3.1.2 Booking a Seat

After selecting the bus passenger can book a seat

Priority level: high

REQ-1	Book a seat
Input	Click the Book your seat button
Process	Load the booking page
Output	Display the fields to enter details
Definition	Taking passenger details to book a seat

REQ-2	Save passenger booking details
Input	Input the necessary details and click submit button
Process	Save all details in database
Output	-
Definition	Storing booking details

3.1.3 Owner Side requirements

Priority level: high

REQ-1	Manage all booking and bus details
Input	Click the Admin button
Process	Load the admin page
Output	Display fields to enter username and password
Definition	Display all details about busses and bookings

3.2 Non-functional Requirements

3.2.1 Performance

- **Responsive Search Results:** The application must deliver responsive search results, ensuring that users receive quick and relevant responses to their queries regarding PG accommodations. This includes optimizing backend processes such as data retrieval and filtering to minimize latency.
- **Map Interactions:** Interactive map functionalities should provide smooth navigation and seamless interaction for users exploring PG accommodation locations. Features such as zooming, panning, and marker interactions should be responsive and intuitive.
- **Compatibility:** Room Scout should be compatible with various mobile platforms (iOS, Android) and screen sizes, ensuring a consistent user experience across different devices. The application's responsiveness and layout adaptation should cater to the diversity of mobile devices in the market.

3.2.2 Security

- **Data Encryption:** User data, including personal information and authentication credentials, will be encrypted using industry-standard encryption algorithms. This ensures that sensitive data stored in the database is protected against unauthorized access and data breaches.
- **Secure Authentication Protocols:** Robust authentication protocols, such as JWT (JSON Web Tokens) for session management and Bcrypt.js for password hashing, will be implemented to safeguard user accounts. This includes mechanisms to prevent common security threats like cross-site scripting (XSS) and cross-site request forgery (CSRF).

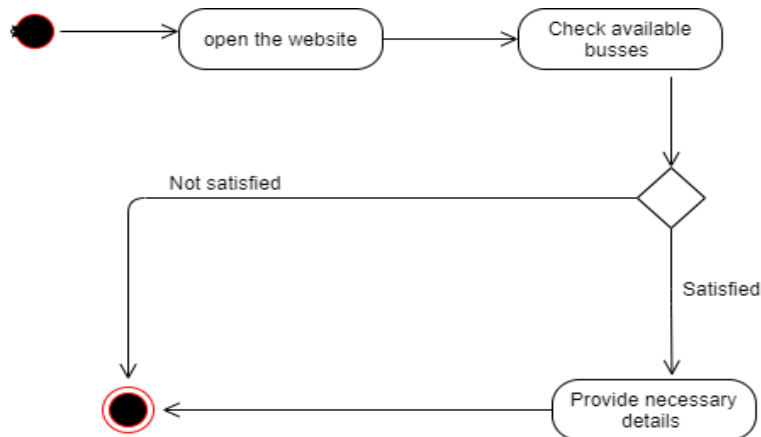
3.2.3 Usability

- **Intuitive User Interface:** The application will feature an intuitive user interface (UI) designed for easy navigation and clear information display. User interactions, from searching for accommodations to managing Wishlist's and leaving reviews, should be straightforward and user-friendly.
- **Third-Party Integration:** Integration with third-party services, particularly for map functionalities, will be seamless and enhance the app's usability. This includes ensuring that map interactions and location-based features are integrated smoothly into the overall user experience without disruptions.

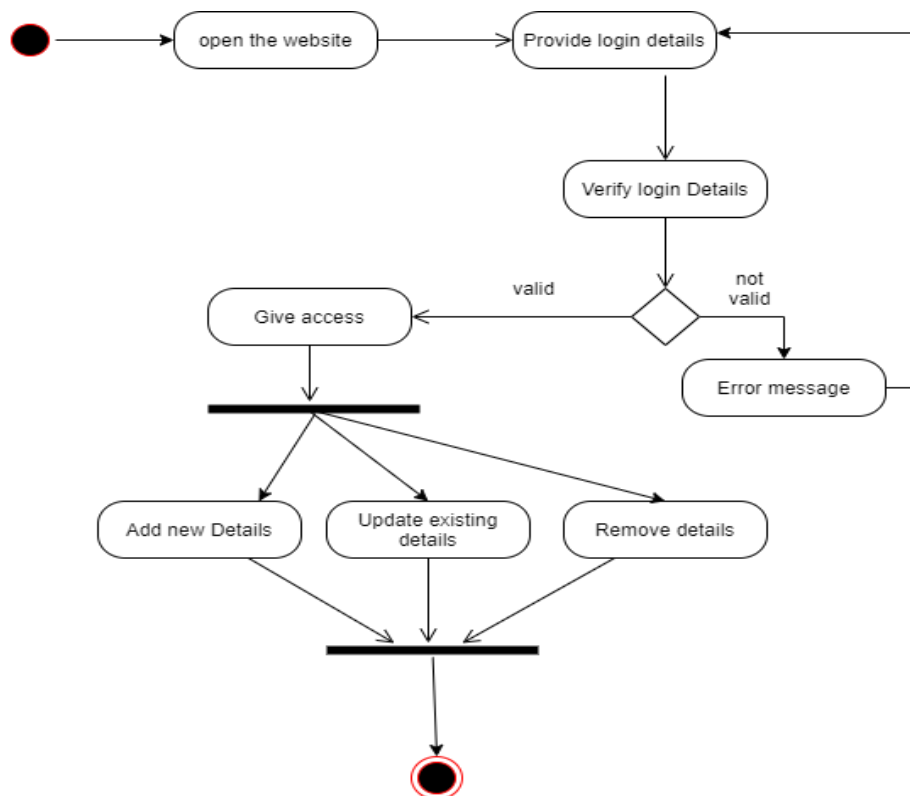
3.3.1 Activity Diagram

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration, and concurrency. The activity diagrams which describe the method of publishing new bus entries by owner and passenger sided overview is given below.

- **Passenger sided activity diagram**



- **Client-side activity**

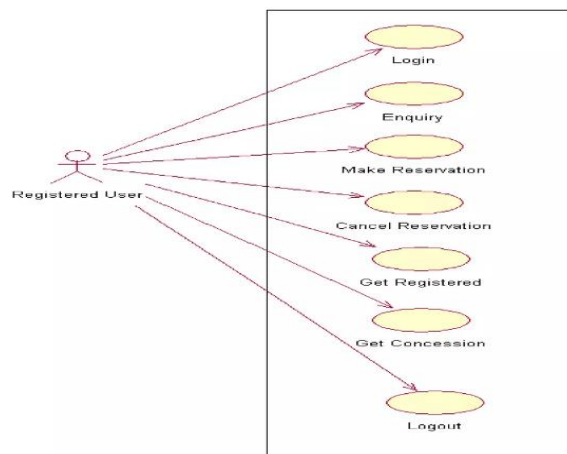


3.3.2 Interfaces of website

Use-Case Diagram

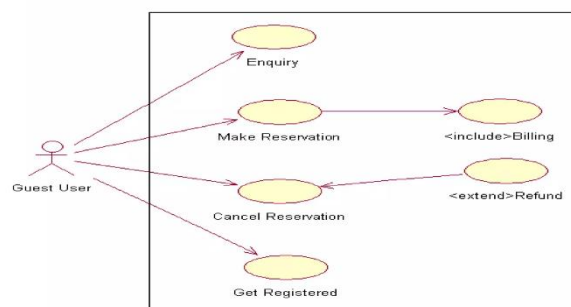
A Use Case Diagram is defined as a graphical representation of the interactions between users (actors) and a system. It illustrates the various ways in which users interact with a system to achieve specific goals or tasks.

USE CASE diagram for registered user:



Use Case Diagram for Registered User

USE CASE diagram for quest user:

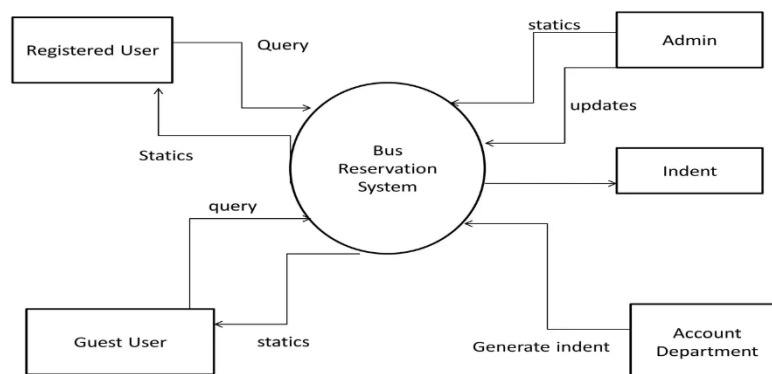


Use Case Diagram for guest User

Data-Flow-Diagram

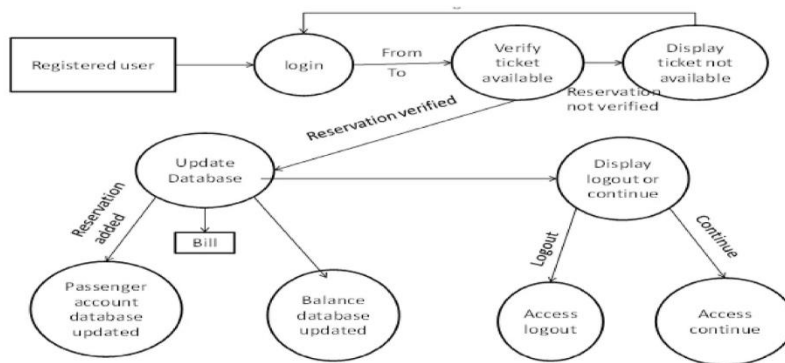
A **Data Flow Diagram (DFD)** is a graphical representation of the "flow" of data through an information system. It can be used to visualize data processing in a structured way, allowing a user or analyst to understand the system and its components briefly. The DFD can map out the flow of information for any process or system, utilizing predefined symbols and notations to represent data inputs, outputs, storage points, and the paths between each destination.

DFD DIAGRAM "0 level":



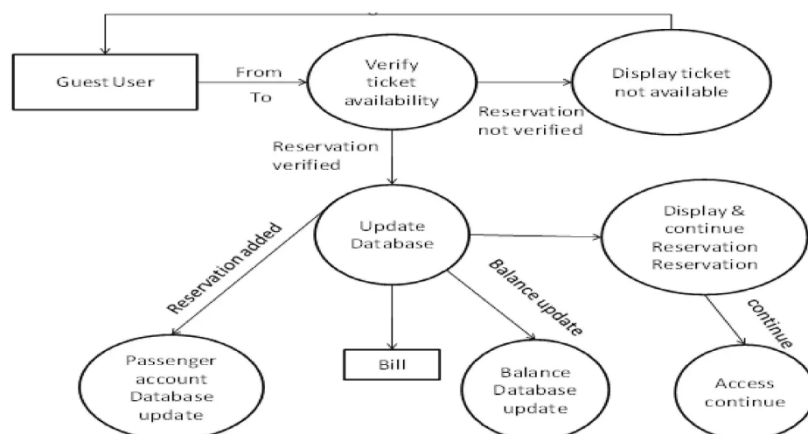
Data Flow Diagram

DFD Diagram for registered user:



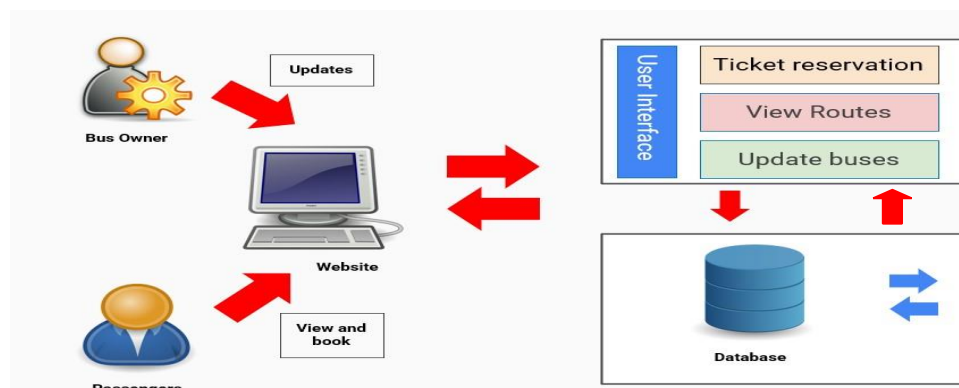
DFD for registered users

DFD for guest user:



DFD for guest user

3.4. IMPLEMENTATION



implementation

3.4.1 Database connection

Connection between the website and the database is established by a separate PHP file called

'db_connect.php'. That specific file is used in every webpage where the database connection is necessary.

3.4.2 Login and logout

Login is only written to the admin of the site. We didn't make a login to the passenger who are booking the bus. Passengers will only want to enter some of their information when booking the bus, so they won't require to login.

In case another one wants to access the system; an admin can only add someone to the system to modify the webpage.

Logging out from the application can be done through the 'Logout' button located at the top right corner of the admin page.

3.4.3 Admin page

Admin has every permission to change any information present in the site. Admin can add, remove or update busses, routes, schedules, cities and also can change the site settings. Admin can add another person as an admin or a staff as he preferred to modify the site.

A staff user cannot change any site settings or add users, but user also have the access to add, remove, or update busses, routes, schedules, and cities.

3.4.4 Search

Searching is one of the core features of this application. A passenger can search the bus routes from a drop-down list and select a date to book a bus. The search results will be displayed in the next screen 'Bus List'.

4.Coding

Introduction:

The goal of coding or programming phase is to translate the system design, produced during the designing phase, into code in each programming language which can be executed by a computer and that performs the computation specified by the design. During the implementation, it should be kept in mind that the programs should not be constructed so that they are easy to write, but that they are easy to read and understand.

4.1 Programming practices:

4.1.1 Top-down & Bottom-up approaches:

In a top-down implementation, the implementation starts from the top of the hierarchy and proceeds to the lower levels. First the main module is implemented, and their subordinates, and so on. In a bottom-up implementation, the process is the reverse. The development starts with implementing the modules at the bottom of the hierarchy and proceeds through the higher levels until it reaches the top. Top-down and bottom-up implementation should not be confused with top-down and bottom-up design, here the design is being implemented, and if the design is fair detailed and complete, its implementation can proceed in either the top-down manner.

4.2.2 Structured Programming:

The basic objective of the coding activity is to produce programs are easy to understand. A program has a static structure as well as a dynamic structure. The static structure is the structure of the text of the program, which is usually just a line organization of statements of the program. The dynamic structure of the program is the sequences of statements executed during the execution of the program. In other words, both the static structure and the dynamic behavior are sequences of statements; when the sequence representing the static structure is fixed, the sequences of statements it executable can change from execution to execution.

4.2.3 Information Hiding:

A software solution to a problem always contains data structures that are meaningfully represent information in the problem domain. That is, when software is developed results a problem, the software uses some data structures to capture the information in the problem domain. Any software solution to a problem contains data structures that represent information in the problem domain. In the problem domain, in general, only certain operations are performed on some information. That is, a piece of information in the problem domain is used only in a limited number of ways in the problem domain.

4.2.4 Programming style:

Here we will list some general rules that can be applied for writing good code.

- **Names:** Selecting module and variable names is often not considered important novice programmers. Most variable sin a program reflects some entity in the problem domain, and the modules reflect some process. Variable names should be closely

related to the entity they represent, and module names should reflect their activity.

- **Control constraints:** As discussed earlier, it is desirable that as much as possible single entry, single exit constructs be used. It is also desirable to use a few standard control constructs rather than using a wide variety of constructs, just because they are available in the language.

- **Information hiding:** As discussed earlier, information hiding should be supported where possible. Only the access functions for the data structures should be made visible while hiding the data structures behind these functions.

- **User-defined types:** Modern languages allow users to define data types when such facilities are available, they should be exploited where applicable.

- **Nesting:** The different control constructs, particularly the if-then-else, can be nested.

- If the nesting becomes too deep, the programs become harder to understand. In case of deeply nested if-then-else, it is often difficult to determine if statement to which a particular else clause is associated. If possible, deep nesting should be avoided.

4.2 Verification:

Verification of the output of the coding phase is primarily interested in detecting errors introduced during this phase. That is, the goal of verification of the code produced is to show that the code is consistent with the design is supposed to implement. It should be avoided out that by verification we mean providing correctness of programs. Program verification method falls into two categories static and dynamic methods. In dynamic methods, the program executes some test data, and the outputs of the program are examined to determine if there are any errors present. Static techniques, on the other hand, do not involve actual program execution on actual numeric data, though it may involve some form of conceptual execution:

4.2.1 Code Reading:

Code reading involves careful reading of the code by the programmer to detect any discrepancies between the design specifications and the actual implementation. It involves determining the abstraction of a module and then comparing it with its specifications. The process of code reading is best done by reading the code inside-out, starting with the innermost structure of the module. First determine its abstract behavior and specify the abstraction. Then the higher-level structure is considered, with the inner structure replaced by its abstraction. This process is continued until we reach the module or program being read.

4.2.2 Static analysis:

Analysis of programs by methodically analyzing the program text is called static analysis is usually performed mechanically by the aid of software tools. During static analysis the program itself is not executed, but the program text is the input to the tools. The aim of the static analysis tools is to detect errors or potential errors or to generate information about the structure of the program that can be useful for documentation or understanding of the programs. An advantage is that static analysis sometimes detects the errors themselves, not just the presence of errors as in testing. This saves the effort of tracing the error from the data that reveals the presence of errors. Static analysis can provide insight into the structure of the program.

4.2.3 Symbolic execution:

Inputs to the program are not numbers but symbols representing the input data, which can take different values. The execution of the program proceeds like normal execution, except that it deals with values that are not numbers but formulates consisting of the symbolic input values. The outputs are symbolic formulae of input values. These formulae can be checked to see if the program will behave as expected. This approach is called as symbolic execution.

4.3 Coding for Admin:

```
<!DOCTYPE html>
<html>
  <head>
    <?php include('header.php') ?>
    <?php
      // session_start();
      // if(isset($_SESSION['login_id'])){
      //   header('Location:home.php');
      // }
    ?>
    <title>Admin Login |Bus Booking</title>
  </head>
  <style>
    body {
      background-image: url(./assets/img/bus.jpg);
      height: 96vh;
      background-position: center;
      background-repeat: no-repeat;
      background-size: cover;
    }
  </style>
  <body id='login-body' class='bg-light">
    <div class="card col-md-4 offset-md-4 mt-4">
      <div class="card-header-edge text-white">
        <strong>Login</strong>
      </div>
      <div class="card-body">
        <form id="login-frm">
          <div class="form-group">
            <label>Username</label>
            <input type="username" name="username" class="form-
control">
          </div>
          <div class="form-group">
```

```

        <label>Password</label>
        <input type="password" name="password" class="form-
control">
    </div>
    <div class="form-group text-right">
        <button class="btn btn-primary btn-block"
name="submit">Login</button>
    </div>

</form>
</div>
</div>

</body>

<script>
$(document).ready(function(){
    $('#login-frm').submit(function(e){
        e.preventDefault()
        $('#login-frm button').attr('disable',true)
        $('#login-frm button').html('Please wait...')

        $.ajax({
            url: './login_auth.php',
            method: 'POST',
            data: $(this).serialize(),
            error: err=>{
                console.log(err)
                alert('An error occured');
                $('#login-frm button').removeAttr('disable')
                $('#login-frm button').html('Login')
            },
            success: function(resp){
                if(resp == 1){
                    location.replace('index.php?page=home')
                }else{
                    alert("Incorrect username or password.")
                    $('#login-frm button').removeAttr('disable')
                    $('#login-frm button').html('Login')
                }
            }
        })
    })
})
</script>

```

</html>

4.3 Coding for Customer Booking:

```
<?php
session_start();
include('db_connect.php');
if(isset($_GET['id']) && !empty($_GET['id'])) {
    $qry = $conn->query("SELECT * FROM schedule_list where id =
".$_GET['id']->fetch_array();
    foreach($qry as $k => $val){
        $meta[$k] = $val;
    }
    $bus = $conn->query("SELECT * FROM bus where id = ".$meta['bus_id']->
    fetch_array();
    $from_location = $conn->query("SELECT id,Concat(terminal_name,',
    ',city,', ',state) as location FROM location where id
    =".$meta['from_location']->fetch_array();
    $to_location = $conn->query("SELECT id,Concat(terminal name,', ',city,',
    ',state) as location FROM location where id =".$meta['to_location']->
    fetch_array();
    $count = $conn->query("SELECT SUM(qty) as sum from booked where
    schedule_id = ".$meta['id']->fetch_array()['sum'];
}
if(isset($_SESSION['login_id']) && isset($_GET['bid'])){
    $booked = $conn->query("SELECT * FROM booked where
    id=".$_GET['bid']->fetch_array();
    foreach($booked as $k => $val){
        $bmeta[$k] = $val;
    }
}
?>
<div class="container-fluid">
    <form id="manage_book">
        <div class="col-md-12">
            <p><b>Bus:</b> <?php echo $bus['bus_number'] . ' |
            '.$bus['name'] ?></p>
            <p><b>From:</b> <?php echo $from_location['location']
            ?></p>
            <p><b>To:</b> <?php echo $to_location['location']
            ?></p>
            <p><b>Departure Time</b>: <?php echo date('M d,Y h:i
            A',strtotime($meta['departure_time'])) ?></p>
            <p><b>Estimated Time of Arrival:</b> <?php echo
            date('M d,Y h:i A',strtotime($meta['eta'])) ?></p>
```

```

        <?php if(($count < $meta['availability']) ||
isset($_SESSION['login_id'])): ?>
            <input type="hidden" class="form-control" id="sid"
name="sid" value='<?php echo isset($_GET['id']) ? $_GET['id'] : " ?>'
required="">
            <input type="hidden" class="form-control" id="sid"
name="bid" value='<?php echo isset($_GET['bid']) ? $_GET['bid'] : " ?>'
required="">

            <div class="form-group mb-2">
                <label for="name" class="control-
label">Name</label>
                <input type="text" class="form-control" id="name"
name="name" value="<?php echo isset($bmeta['name']) ? $bmeta['name'] : "
?>">
            </div>
            <div class="form-group mb-2">
                <label for="qty" class="control-
label">Quantity</label>
                <input type="number" maxlength="4" class="form-
control text-right" id="qty" name="qty" value="<?php echo
isset($bmeta['qty']) ? $bmeta['qty'] : " ?>">
            </div>
            <?php if(isset($_SESSION['login_id'])): ?>
            <div class="form-group mb-2">
                <label for="qty" class="control-
label">Status</label>
                <select class="form-control" id="status"
name="status" value="<?php echo isset($bmeta['qty']) ? $bmeta['qty'] : "
?>">
                    <option value="1" <?php echo
isset($bmeta['status']) && $bmeta['status'] == 1 ? "selected" : "
?>>Paid</option>
                    <option value="0" <?php echo
isset($bmeta['status']) && $bmeta['status'] == 0 ? "selected" : "
?>>Unpaid</option>
                </select>
            </div>
            <?php endif; ?>
            <?php else: ?>
            <h3>No Available seat</h3>
            <style>
                .uni_modal .modal-footer{
                    display: none;
                }
            </style>

```

```

        <?php endif; ?>
    </div>
</form>
</div>

<script>
    $('#manage_book').submit(function(e){
        e.preventDefault()
        start_load()
        $.ajax({
            url: './book_now.php',
            method: 'POST',
            data: $(this).serialize(),
            error: err=>{
                console.log(err)
            },
            end_load()
            alert_toast('An error occurred', 'danger');
        },
        success: function(resp){
            resp = JSON.parse(resp)
            if(resp.status == 1){
                end_load()
                $('#modal').modal('hide')
                alert_toast('Data successfully saved', 'success');
                if('<?php echo !isset($_SESSION['login_id']) ?>'
== 1){
                    $('#book_modal .modal-body').html('<div
class="text-
center"><p><strong><h3>'+resp.ref+'</h3></strong></p><small>Referenc
e Number</small><br/><small>Copy or Capture your Reference number
</small></div>')
                $('#book_modal').modal('show')
            }else{
                load_booked();
            }
        }
    })
    $('#datetimepicker').datetimepicker({
        format: 'Y/m/d H:i',
        startDate: '+3d'
    });
</script>

```

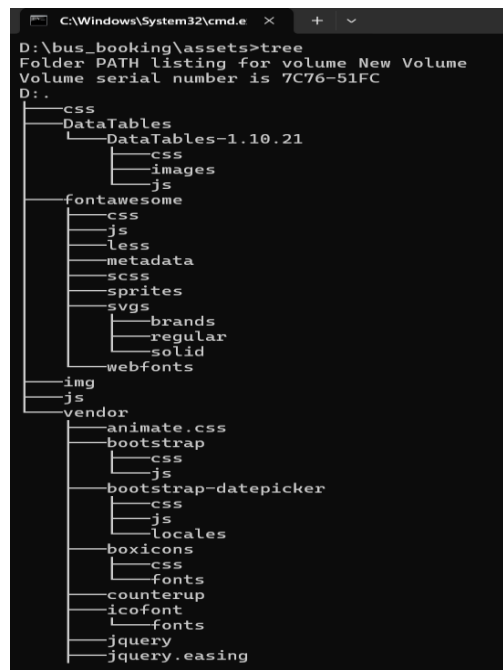
5. Working

Back-End

5.1 Directory Structure

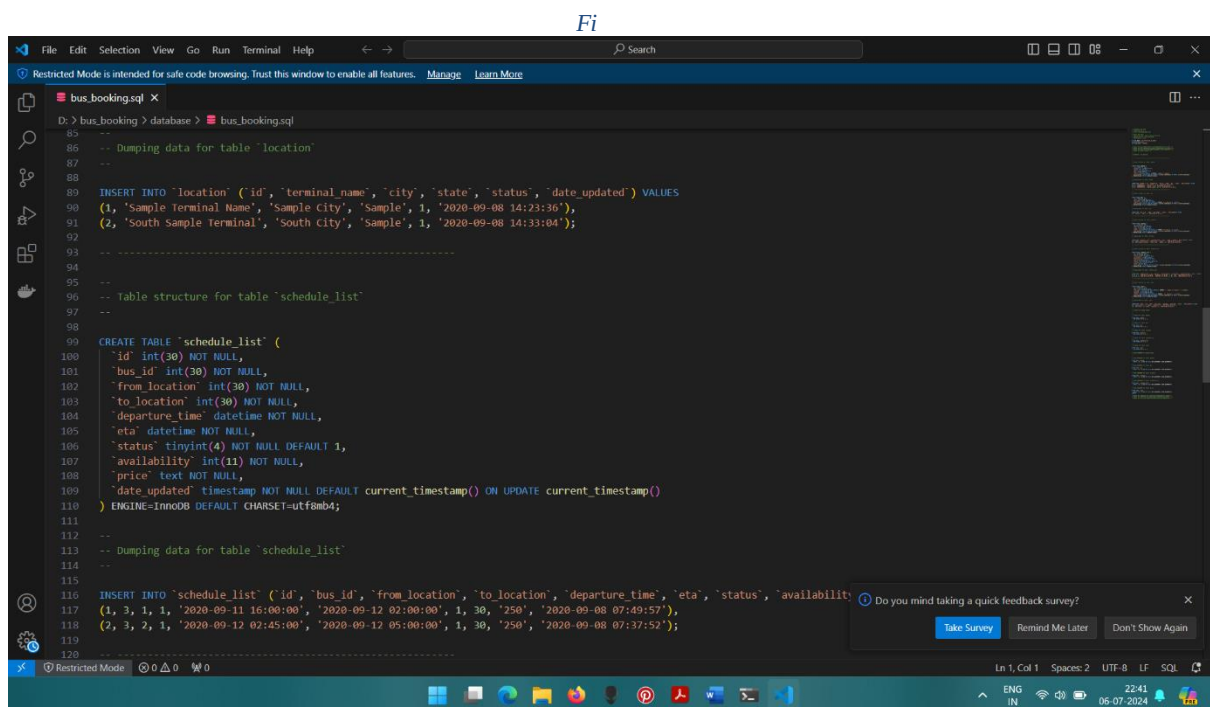
```
D:\bus_booking>TREE
Folder PATH listing for volume New Volume
Volume serial number is 7C76-51FC
D: .
├── assets
│   ├── css
│   ├── DataTables
│   │   └── DataTables-1.10.21
│   │       ├── css
│   │       ├── images
│   │       └── js
│   ├── fontawesome
│   │   ├── css
│   │   ├── js
│   │   ├── less
│   │   ├── metadata
│   │   ├── scss
│   │   ├── sprites
│   │   ├── svgs
│   │   │   ├── brands
│   │   │   ├── regular
│   │   │   └── solid
│   │   └── webfonts
│   ├── img
│   ├── js
│   └── vendor
│       ├── animate.css
│       ├── bootstrap
│       │   ├── css
│       │   └── js
│       ├── bootstrap-datepicker
│       │   ├── css
│       │   ├── js
│       │   └── locales
│       ├── boxicons
│       │   ├── css
│       │   └── fonts
│       ├── counterup
│       ├── icofont
│       │   └── fonts
│       ├── jquery
│       ├── jquery.easing
│       ├── owl.carousel
│       │   └── assets
│       ├── php-email-form
│       ├── remixicon
│       ├── venobox
│       └── waypoints
├── database
└── forms
```

5.2 Directory Structure



5.3 Bus_Booking.sql (Database file)

Creating tables for the various modules in the project such as schedule list, booked, bus, location.



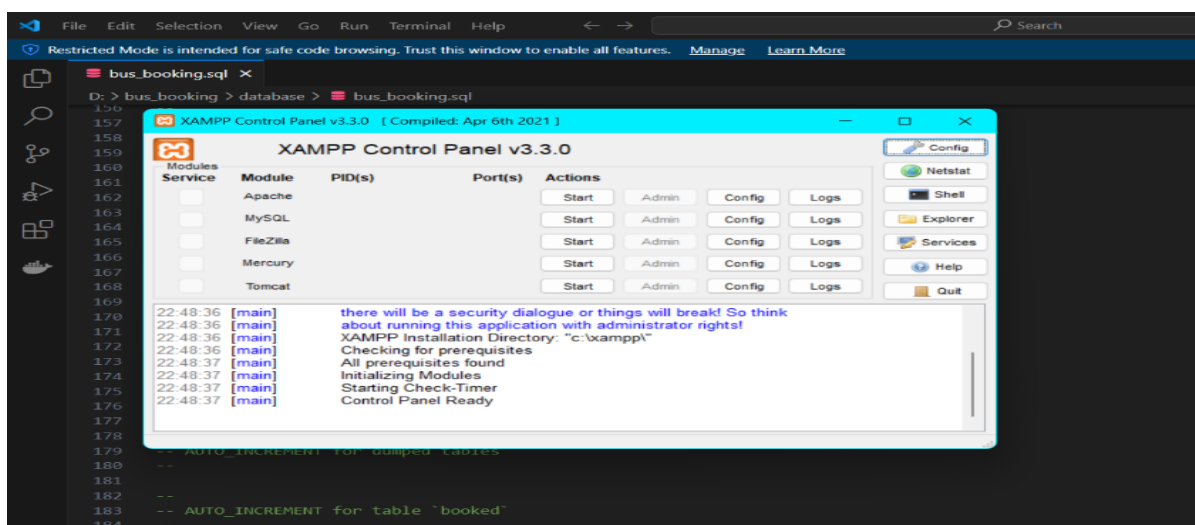
Database file

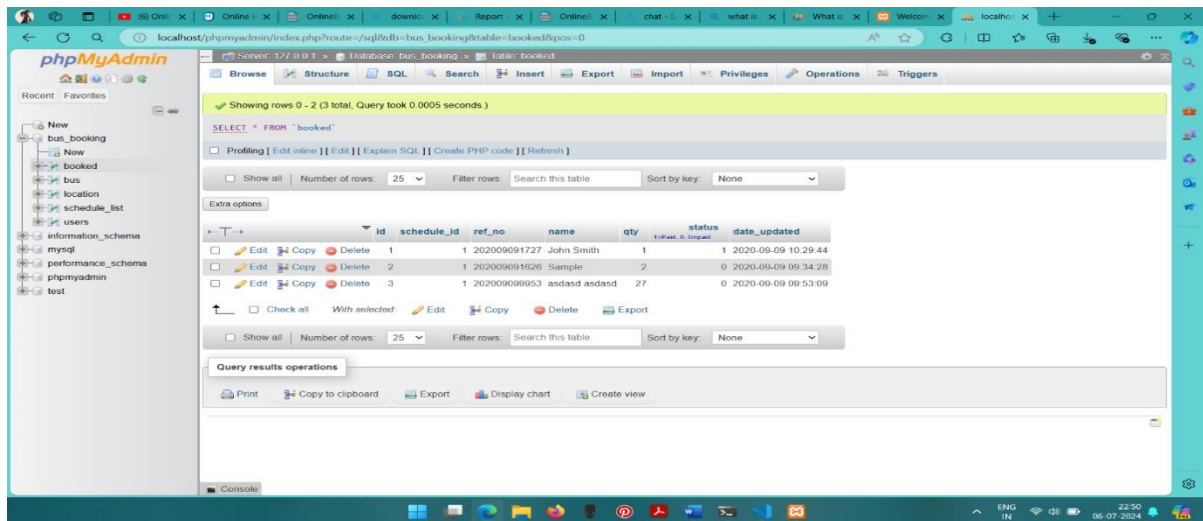
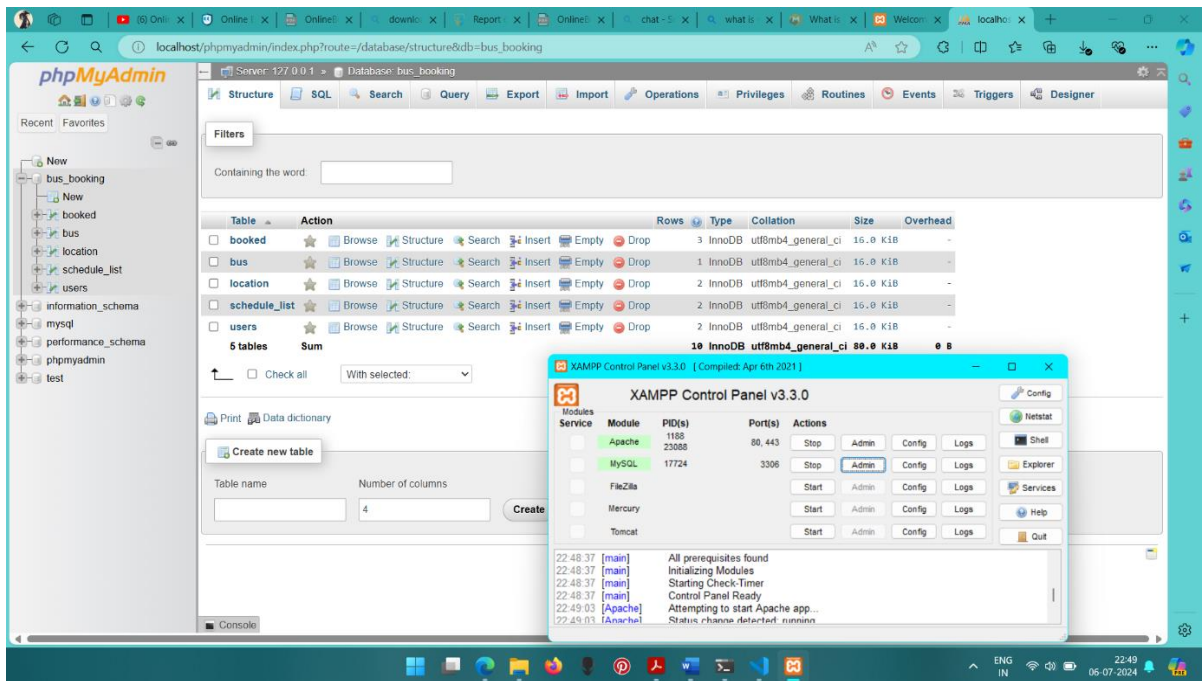
```
D: > bus_booking > database > bus_booking.sql
-- Table structure for table 'booked'
--
--
CREATE TABLE `booked` (
  `id` int(30) NOT NULL,
  `schedule_id` int(30) NOT NULL,
  `ref_no` text NOT NULL,
  `name` varchar(250) NOT NULL,
  `qty` int(11) NOT NULL,
  `status` tinyint(1) DEFAULT 0 COMMENT '1=Paid, 0= Unpaid',
  `date_updated` datetime NOT NULL DEFAULT current_timestamp() ON UPDATE current_timestamp()
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

--
-- Dumping data for table 'booked'
--
INSERT INTO `booked` (`id`, `schedule_id`, `ref_no`, `name`, `qty`, `status`, `date_updated`) VALUES
(1, 1, '202009091727', 'John Smith', 1, 1, '2020-09-09 10:29:44'),
(2, 1, '202009091626', 'Sample', 2, 0, '2020-09-09 09:34:28'),
(3, 1, '202009099953', 'asdasd asdasd', 27, 0, '2020-09-09 09:53:09');

--
-- Table structure for table 'bus'
--
--
CREATE TABLE `bus` (
  `id` int(30) NOT NULL,
  `name` varchar(250) NOT NULL,
  `bus_number` varchar(50) NOT NULL,
  `status` tinyint(1) NOT NULL DEFAULT 1 COMMENT '0 = inactive, 1 = active',
  `date_updated` datetime NOT NULL DEFAULT current_timestamp() ON UPDATE current_timestamp()
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

XAMPP STRUCTURE AND WORKING





Databases inside XAMPP

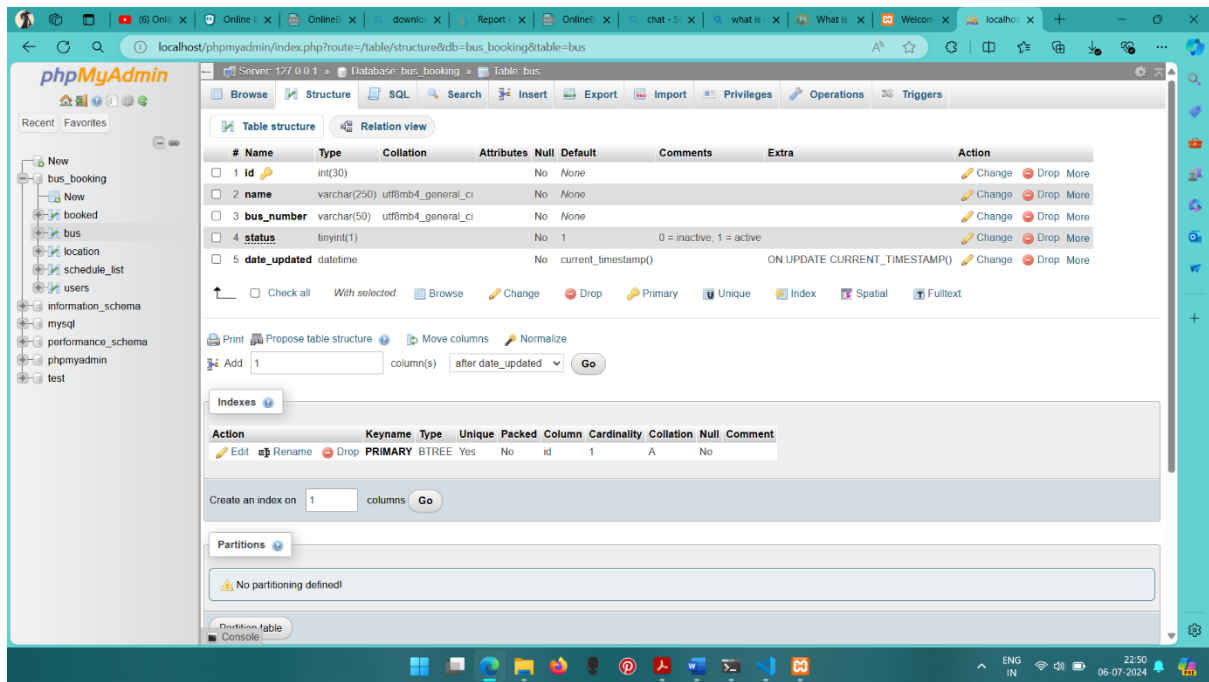
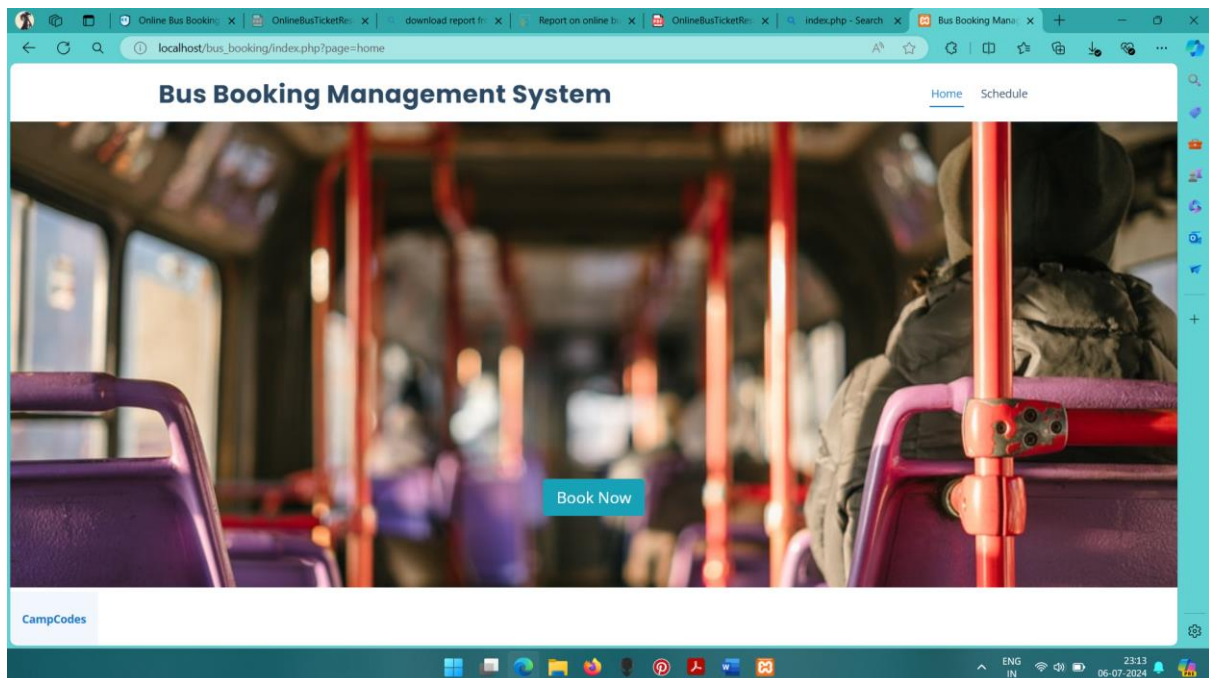


table for bus module

Output

Front-End



Home Page

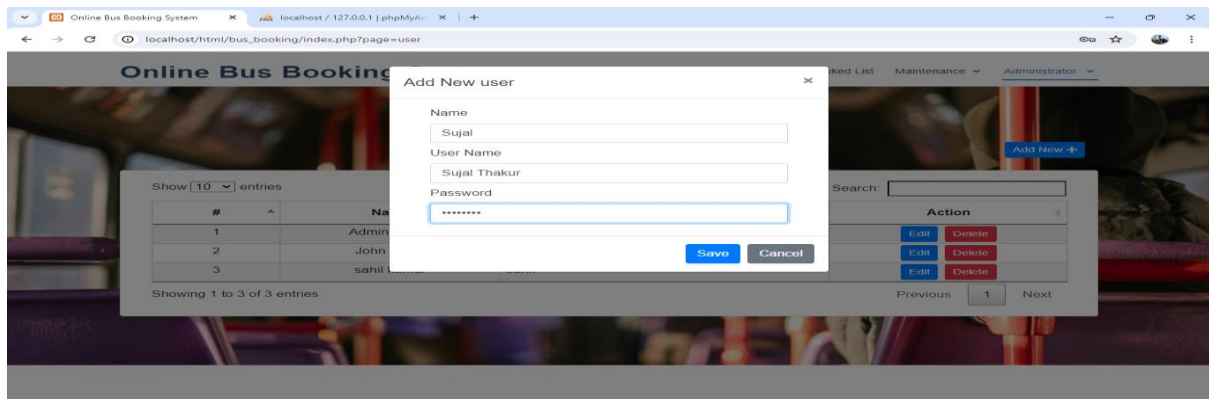
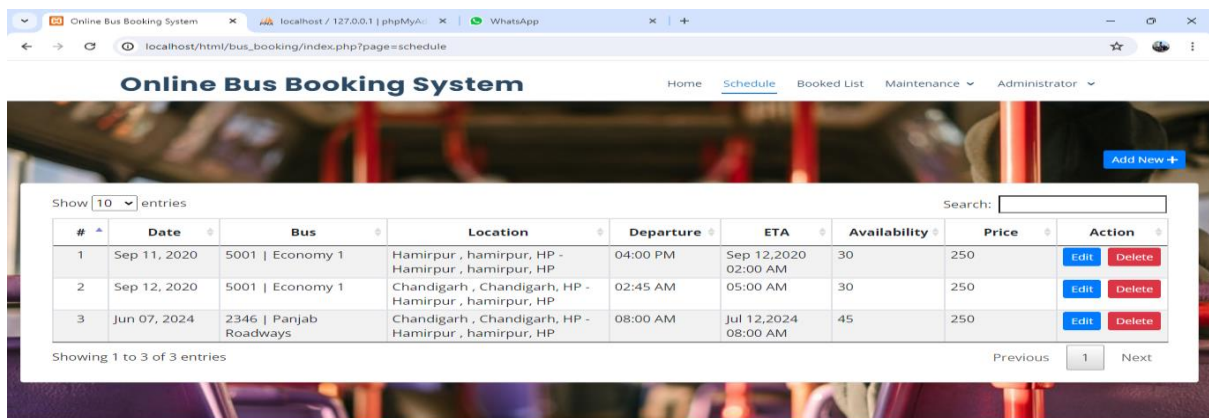
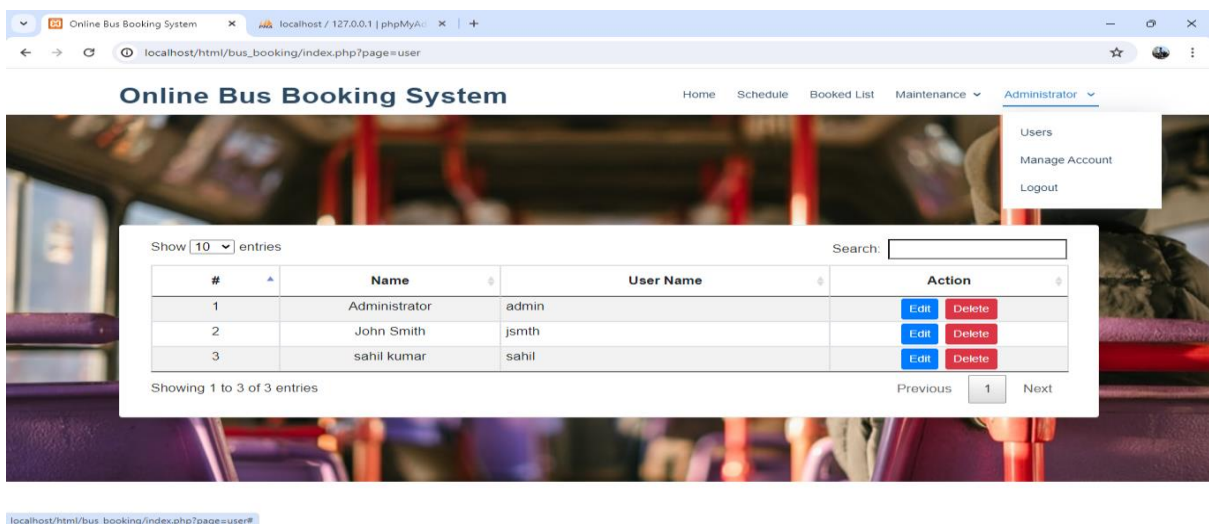


Figure 3 admin page: add new user



schedule for available buses



user's login information

Conclusion

Assessment of the project

The goal of the project was to provide a ticket booking method for passengers. Our project was online bus ticket booking system it provides users to book their bus tickets via online. Due to this COVID pandemic situation, transportation has several restrictions. Due to prevailing COVID situation government had decided to establish the transportation rules & regulations. Considering about that scenario We made a web application for ticket booking purpose. As we said before in this application passengers can book a ticket, update, or delete it. And bus owners can add new buses & routes.

Our project is totally based on web applications. HTML, CSS, Php, MYSQL are the major factors of our project. As a team we decided to execute our project as web application because websites act a major role in business purposes and marketing purposes as well. It's the backbone of a business. Eye catching visual interactions, high responsiveness is mainly considered in web development. Because as we are in digitalized era whole community likes to accomplish their task as much as easier. For that reason, different types of web-based applications and websites interfered with users to achieve their tasks. Websites is a good method for reaching larger audience as an example we created an online bus ticket reservation application it will useful for large number of customers at short time of period. Also, all the processes going to execute online that is an additional advantage for users. Our webpage's home page consists clear letters with larger font size and recognizable symbols, commands with navigation bar to provide the users a friendly GUI. Users can check the available buses for a specific route, book a bus are some functionalities of our webpage. Before the Final outcome of the project, we had to accomplish five main milestones.

- a) Project charter- consists brief idea of our project idea.
- b) Project proposal – to depicts the in-detail concepts and the plan of the project.
- c) SRS submission – contains the system's purpose, functionalities & project roles.
- d) Project progress – explains how far the progress of the ongoing project.
- e) Final evaluation - describes the outcome of the project.

Considering about the performances of our project, users can book a specific route

and, they can delete & update existing bookings. As an additional option bus owners can enter their bus details for this web application. All the details are managed in a database which can access only from administers side.

Lesson learnt

When doing this project, we mainly consider about the usability of this application. Due to these pandemic crisis, large number of resources was offered via online methods. As we are expected, transportation services also adhere to that. After the discussion of our team members, we decided to execute our project idea. This application used different programming languages and concepts in project initialization procedures. While building this outcome we were able to identify our teammates strengths and weaknesses also. Our main goal was to create a platform for both bus owners and passengers for their travelling purposes. We think we were able to accomplish it.

Future work

An organization should have an income to further development of products. Advertising of project is an essential fact. Because there are certain number of tickets booking application currently also. We must specially concern about the quality of our application to compete with other developers. We must clearly express our QOS for stakeholders. Minimizing the software errors and enhancing system procedures can ensure the quality of an application. Furthermore, mobile version of this application will be grateful for future progression for this project. Because mobile phone is more personalized, more user-friendly, and brand presence & faster. Most of the organizations care about the customer's satisfaction, to guaranteed that we had to improve our system's features also. Although as a business we highly expect the growth of a business. For accomplish that goals future work plans and procedures are essential.